Financial Fraud and Money-Laundering Detection in Transaction Data

COP 3045 Python Programming II - Team 2 | October 11, 2026

PaySim (6,362,620 mobile-money transactions) and IBM AML HI-Small (5,078,345 bank transactions), both synthetic.

1. Dataset Summary

Total observations: 11,440,965

Total variables: 27 (union of both datasets' columns)

Not suspicious transactions: 11,427,575

Suspicious transactions: 13,390

Overall suspicious rate: 0.1170%

Records by dataset:

PaySim: 6,362,620

IBM: 5,078,345

Label types:

fraud: 6,362,620

laundering: 5,078,345

2. Summary Statistics

PaySim amounts have no currency (N/A). IBM amounts use its most common currency (US Dollar).

Dataset	Currency	Count	Mean	Median	Minimum	Maximum
IBM	US DOLLAR	1,879,341	313,640.22	884.04	0.01	16,620,608,535.74
PaySim	N/A	6,362,620	179,861.90	74,871.94	0.00	92,445,516.64

Spread and quartiles:

Dataset	Currency	Std. Dev.	25th Percentile	75th Percentile
IBM	US DOLLAR	17,454,909.53	157.95	5,147.06
PaySim	N/A	603,858.23	13,389.57	208,721.48

The IBM maximum of 16,620,608,535.74 is an extreme outlier, so the median is the more reliable measure of a typical transaction.

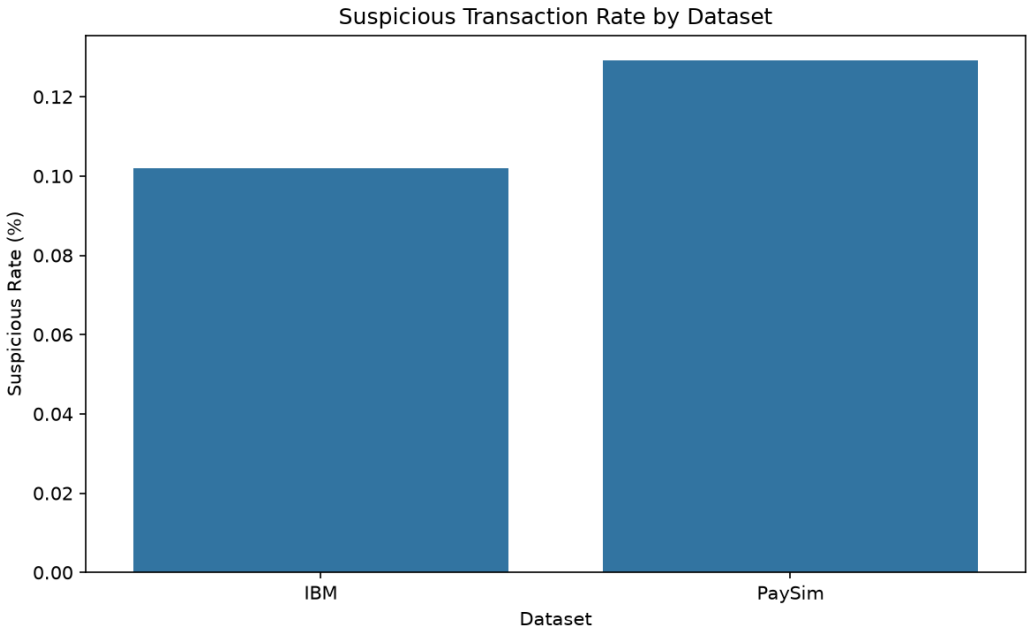
Categorical summary (top 8 categories by suspicious rate):

Counts provide context for the rate; categories with few transactions can have volatile rates.

Datase			
PaySim - TRANSFER	532,909	0.7688%	
IBM - ACH	600,797	0.7462%	
PaySim - CASH_OUT	2,237,500	0.1840%	
IBM - BITCOIN	146,091	0.0383%	
IBM - CASH	490,891	0.0220%	
IBM - CHEQUE	1,864,331	0.0174%	
IBM - CREDIT CARD	1,323,324	0.0156%	
IBM - REINVESTMENT	481,056	0.0000%	

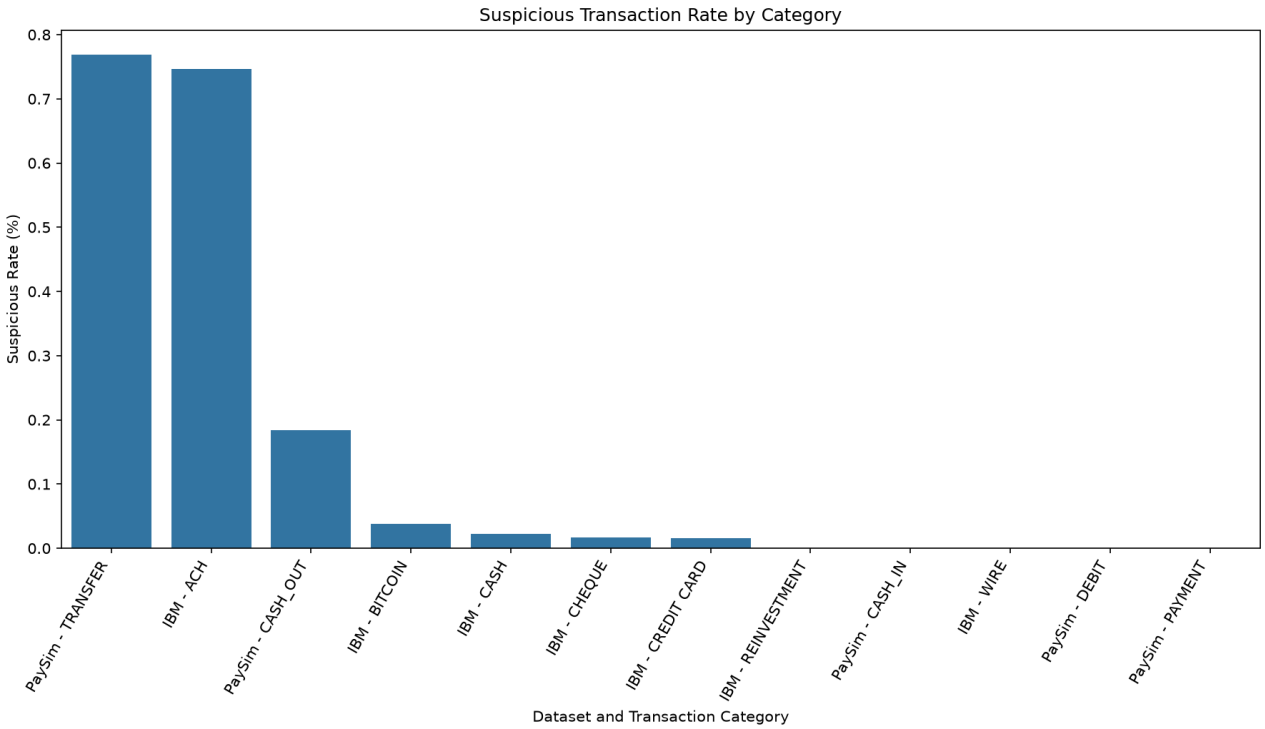
3. Visualizations

Suspicious Transaction Rate by Dataset



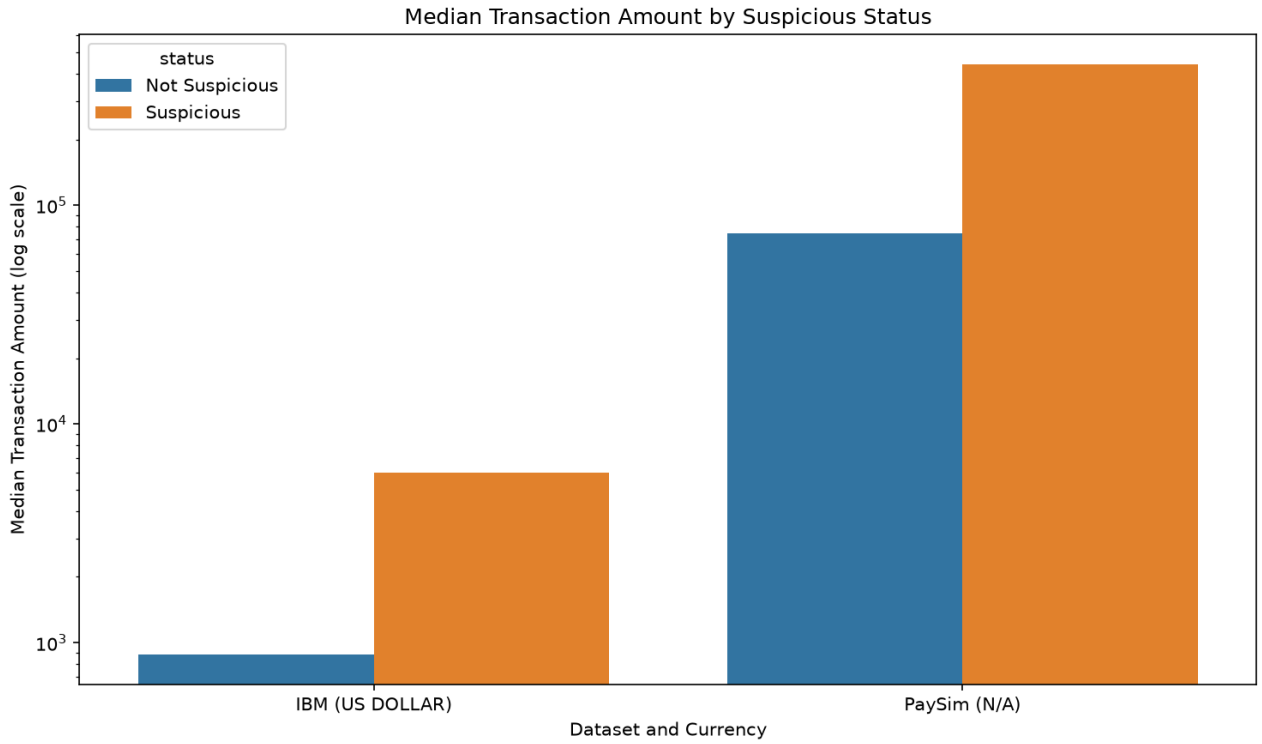
Suspicious rates are shown separately for PaySim fraud and IBM laundering labels; interpret each relative to its dataset.

Suspicious Transaction Rate by Category



Categories are ordered by suspicious rate; category rates should be read alongside their transaction counts.

Median Transaction Amount by Suspicious Status



The logarithmic vertical axis keeps the PaySim and IBM medians readable on the same chart.

4. Key Analytical Findings

The combined analytical dataset contains 11,440,965 transactions and 27 variables.

There were 13,390 suspicious transactions, giving an overall suspicious rate of 0.1170%.

PaySim had the highest suspicious transaction rate among the datasets at 0.1291%.

The transaction category with the highest suspicious rate was TRANSFER from PaySim, with a suspicious rate of 0.7688%.

In IBM (US DOLLAR), the median suspicious transaction (5,970.08) was 6.8x the median non-suspicious one (881.72).

In PaySim (N/A), the median suspicious transaction (441,423.44) was 5.9x the median non-suspicious one (74,684.72).

The three highest-risk categories were PaySim - TRANSFER, IBM - ACH, PaySim - CASH_OUT.

IBM ACH transactions had a suspicious rate of 0.75%.

Limitation: the data are synthetic and highly imbalanced (about 0.12% suspicious), so rates should be read as relative, not as real-world fraud prevalence.